

● HARD

● ADVANCED MATH

---

# Advanced Math

10 questions · ~15 min

# 14

---

SAT QUESTION BANK

2026-05-29

**Q1**

ADVANCED MATH

$$-9x^2 + 30x + c = 0$$

In the given equation,  $c$  is a constant. The equation has exactly one solution. What is the value of  $c$ ?

- A. 3
- B. 0
- C. -25
- D. -53

$$Pt = 2601.04^{\frac{6}{n}t}$$

The function  $P$  models the population, in thousands, of a certain city  $t$  years after 2003. According to the model, the population is predicted to increase by 4% every  $n$  months. What is the value of  $n$ ?

- A. 8
- B. 12
- C. 18
- D. 72

$$\frac{14x}{7y} = 2\sqrt{w+19}$$

The given equation relates the distinct positive real numbers  $w$ ,  $x$ , and  $y$ . Which equation correctly expresses  $w$  in terms of  $x$  and  $y$ ?

A.  $w = \sqrt{\frac{x}{y}} - 19$

B.  $w = \sqrt{\frac{28x}{14y}} - 19$

C.  $w = \frac{x^2}{y} - 19$

D.  $w = \frac{28x^2}{14y} - 19$

$$p(t) = 90,000(1.06)^t$$

The given function  $p$  models the population of Lowell  $t$  years after a census. Which of the following functions best models the population of Lowell  $m$  months after the census?

A.  $r(m) = \frac{90,000}{12}(1.06)^m$

B.  $r(m) = 90,000\frac{(1.06)^m}{12}$

C.  $r(m) = 90,000\frac{(1.06)^{\frac{m}{12}}}{12}$

D.  $r(m) = 90,000(1.06)^{\frac{m}{12}}$

Q5

GRID-IN

ADVANCED MATH

$$-16x^2 - 8x + c = 0$$

In the given equation,  $c$  is a constant. The equation has exactly one solution. What is the value of  $c$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

At the time that an article was first featured on the home page of a news website, there were 40 comments on the article. An exponential model estimates that at the end of each hour after the article was first featured on the home page, the number of comments on the article had increased by 190% of the number of comments on the article at the end of the previous hour. Which of the following equations best represents this model, where  $C$  is the estimated number of comments on the article  $t$  hours after the article was first featured on the home page and  $t \leq 4$ ?

A.  $C = 401.19^t$

B.  $C = 401.9^t$

C.  $C = 4019^t$

D.  $C = 402.9^t$

$$x - 1^2 = -4$$

How many distinct real solutions does the given equation have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

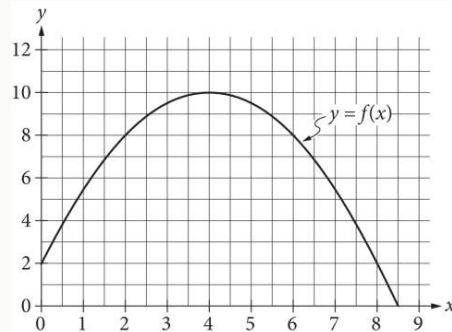


Function  $f$  is a quadratic function where  $f(-20) = 0$  and  $f(-4) = 0$ . The graph of  $y = f(x)$  in the  $xy$ -plane has a vertex at  $(r, -64)$ . What is the value of  $r$ ?

STUDENT-PRODUCED RESPONSE

|   |   |   |   |
|---|---|---|---|
|   |   |   |   |
| . | / | . | . |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



The graph of the function  $f$ , defined by  $f(x) = -\frac{1}{2}(x-4)^2 + 10$ , is shown in the  $xy$ -plane above. If the function  $g$  (not shown) is defined by  $g(x) = -x + 10$ , what is one possible value of  $a$  such that  $f(a) = g(a)$ ?

STUDENT-PRODUCED RESPONSE

|                       |                       |                       |                       |
|-----------------------|-----------------------|-----------------------|-----------------------|
| <input type="text"/>  | <input type="text"/>  | <input type="text"/>  | <input type="text"/>  |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |
| <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Function  $f$  is defined by  $fx = -a^x + b$ , where  $a$  and  $b$  are constants. In the  $xy$ -plane, the graph of  $y = fx - 15$  has a  $y$ -intercept at  $0, -\frac{99}{7}$ . The product of  $a$  and  $b$  is  $\frac{65}{7}$ . What is the value of  $a$ ?

## STUDENT-PRODUCED RESPONSE

|   |    |    |   |
|---|----|----|---|
|   |    |    |   |
| . | /. | /. | . |
| 0 | 0  | 0  | 0 |
| 1 | 1  | 1  | 1 |
| 2 | 2  | 2  | 2 |
| 3 | 3  | 3  | 3 |
| 4 | 4  | 4  | 4 |
| 5 | 5  | 5  | 5 |
| 6 | 6  | 6  | 6 |
| 7 | 7  | 7  | 7 |
| 8 | 8  | 8  | 8 |
| 9 | 9  | 9  | 9 |

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.